

# The AI Usage Spectrum

Five archetypes for how a professional development team works with AI, ordered by what the agent's output can reach with no human action in between. The documents describe rather than recommend: each archetype ends with the posture that fits it, and none is the goal state.

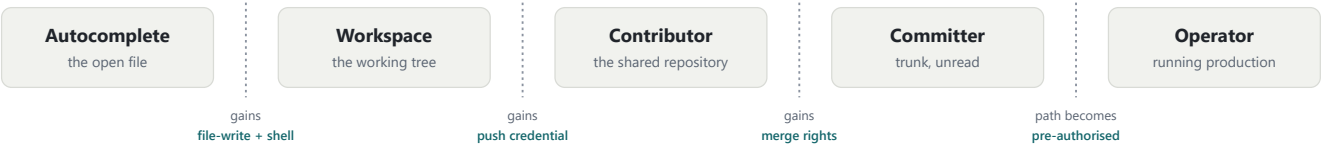
|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| <b>The spectrum</b> | the five archetypes at a glance                           |
| <b>Autocomplete</b> | the open file; a human accepts every edit                 |
| <b>Workspace</b>    | the working tree, locally; nothing leaves without a human |
| <b>Contributor</b>  | the shared repository; a human merges every change        |
| <b>Committer</b>    | trunk, unread; a human still gates release                |
| <b>Operator</b>     | running production; the path is pre-authorised            |
| <b>Glossary</b>     | the ruling vocabulary these documents are written in      |

# Five archetypes, ordered by reach

An archetype is a team's configured ceiling for model-generated changes — what the settings and policies permit, not what any individual change receives. The spectrum orders archetypes by what the agent's output can reach with no human action in between, and each boundary is a capability the previous archetype structurally lacks. Each archetype's summary card follows; each links to its full, cited document.

Vocabulary as of 2026-08-31   Taxonomy: ADR-0004   Sources last verified 2026-09-02

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state. Moving further along the spectrum means replacing human-in-the-loop verification with *engineered* verification — more rigor, not less. Human oversight varies independently within every archetype: every document below states separately what the agent can reach, what a human still gates, and what verification must replace them.

## Autocomplete

reaches the open file

- MADE OF** no tool surface — the model proposes ghost text into the open file, and a human commits every edit with a keypress. The only archetype whose ceiling is architectural rather than configured
- DEMANDS FIRST** nothing new to build — the demand is the discipline of staying here deliberately, one keypress from the surfaces that act

### WHAT IT GAINS YOU

- Every line is read by the person shipping it, at the moment it appears.
- The only region with clean randomised evidence in its favour — and the only one where data egress can be eliminated outright rather than managed.

### WHAT IT COSTS YOU

- The smallest delta on the spectrum: the human still does everything but type.
- Reading becomes a continuous stream of micro-decisions whose acceptance rate measures satisfaction, not output.

*Fits a team that believes reading every line is non-negotiable, and that anything reducing what a human reads costs more than it saves.*

## Workspace

reaches the working tree

**MADE OF** file-write and shell tool surfaces, no push credential — the agent acts across the working tree, locally; nothing leaves the machine without a human

**DEMANDS FIRST** containment: a sandbox, an egress allowlist, least-privilege tool scopes — which the tool field mostly does not ship by default

### WHAT IT GAINS YOU

- Delegated implementation with real feedback loops — the agent edits, builds, runs and tests its own work; one human can direct several agents in parallel.
- A bounded worst case, *if the bounds are built*: local, and largely revertible.

### WHAT IT COSTS YOU

- Containment becomes load-bearing exactly where the field invests least.
- The lethal trifecta — private data, untrusted content, external communication — assembles here by default unless engineered away.

*Fits a team that believes risk is bounded by what the agent can reach, not by who is watching it.*

[Read the full document →](#)

## Contributor

reaches the shared repository

**MADE OF** a push credential — the agent opens pull requests under its own identity; a human merges every change. The archetype the tool market ships and the platform enforces

**DEMANDS FIRST** protection of the review: server-side gates, invitation-only PR creation, task-eligibility classifiers, retry caps — reviewer attention treated as an engineered budget

### WHAT IT GAINS YOU

- Authoring at volume with the team's merge discipline unchanged — agent work comes through the same front door as anyone else's.
- The cheapest ceiling to hold: the boundary above is platform-enforced, so staying here means not building a bypass.

### WHAT IT COSTS YOU

- The review queue becomes the bottleneck, measurably.
- Reviewer habituation is measured and worsens with the volume this archetype itself generates — the safety belief eroding under its own output.

*Fits a team that believes review is the load-bearing control and should stay human — and that an agent's work should come through the same front door as anyone else's.*

[Read the full document →](#)

## Committer

reaches trunk, unread

|                      |                                                                                                                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MADE OF</b>       | merge rights for a defined change class, identified mechanically — author identity, title prefix, severity, risk score — which reaches trunk with no human reading it; a human still gates release |
| <b>DEMANDS FIRST</b> | the qualifying line rendered in code, anti-suppression on the checks, gates enforced server-side, and revert machinery for when the line is wrong                                                  |

### WHAT IT GAINS YOU

- Mechanically-checkable classes stop queueing behind human availability; review attention concentrates where a human can actually help.
- An auditable trust model — every occupant's policy is an inspectable artifact, a workflow file rather than a habit.

### WHAT IT COSTS YOU

- The qualifying line must be engineered, argued for, and policed forever.
- Verification is load-bearing the moment no one reads the diff — and no study anywhere tests unread agent merges against reviewed ones.

*Fits a team that believes trust should be earned by change class rather than by author — and that where a class can be checked mechanically, a human reading it adds cost without adding safety.*

[Read the full document →](#)

## Operator

reaches running production

|                      |                                                                                                                                                                                                     |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MADE OF</b>       | a pre-authorised path to production — nothing human stands between the agent's change and running prod, whether continuous deployment carries an unread merge or the agent holds deploy credentials |
| <b>DEMANDS FIRST</b> | the full engineered chain — gates, canaries, auto-revert, provenance — trusted <i>before</i> the path opens; almost none of it exists wired to autonomy anywhere                                    |

#### WHAT IT GAINS YOU

- Full pipeline speed for the change classes that qualify.
- The clearest possible safety case: stated in machinery, not in habit.

#### WHAT IT COSTS YOU

- First-of-kind wiring, built and policed by you; liability keys to the setting.
- Observed as no team's ceiling — reached per change-class, by narrowing what qualifies, never by widening who may act.

*Fits a team that believes engineered verification can be made a stronger check than a human in the path — and that the honest route there is to narrow what qualifies, never to widen who may act.*

[Read the full document →](#)

Every claim in the five documents carries a citation chip with its source, date, and evidence tier — hard survey data, controlled study or verified primary artifact, vendor-reported metric, or practitioner account — stated with the claim itself. The glossary holds the ruling vocabulary these pages are written in.

# Autocomplete

The agent's output reaches the open file and nothing else. A human accepts every edit with a keypress, and nothing else about the process changes. The fixed low extreme of the spectrum — and the only archetype whose defining property is architectural rather than configured.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Sources last verified 2026-09-02



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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## 01 What this archetype is

An Autocomplete ceiling permits model-generated text to reach **the open file, and nothing else** — and only through a human keypress. The surface is a decoration in an editor buffer: a candidate string is drawn as ghost text, Tab commits it, continued typing discards it. Nothing becomes an edit until the developer commits it while looking at it [• Tooling: Inline & Chat, Aug 2026](#).

Concretely, the step the human stops doing is **typing the characters — and nothing else**. The developer still decides what to build, still reads what appears, still decides whether it is right, and still runs it. Reading is not removed; it is moved earlier and made continuous. This archetype does markedly less than its reputation: the widely-quoted figures — 50% of code characters completed by AI at Google, "*an average of 46% of a developers' code*" from GitHub — describe *character provenance*, not delegated decisions [• Google, Jun 2024](#) [• GitHub, Feb 2023](#). No decision changes hands. That is why the owner fixed this as the spectrum's low extreme: inline AI edits, with nothing else about the process changing [• ADR-0004](#).

## 02 The boundary crossed

None — this is the fixed low extreme. The boundary that defines Autocomplete is the one **not** crossed: **propose versus act**. Across fourteen completion products surveyed, not one documents a completion engine that executes a command, writes to a file the developer is not editing, or makes any outbound call beyond the inference request itself. The architecture forecloses it: a completion engine has no tool surface, so there is nothing to grant [● Tooling: Inline & Chat, Aug 2026](#).

That crossing point is one keypress wide and has a street address: VS Code's terminal inline chat offers a **Run (⌘Enter)** control that executes the suggested command. Everything on the completion side of that control proposes text; everything past it can act [● VS Code docs, Aug 2026](#). And the two sides differ in kind, not degree: **completion acceptance cannot be batched or automated anywhere in the surveyed field** — no vendor ships an auto-accept for ghost text — while in-editor *edit* acceptance acquires an off switch the moment a team crosses over (VS Code's `chat.editing.autoAcceptDelay` converts diff review into a countdown timer) [● Tooling: Inline & Chat, Aug 2026](#). Autocomplete is therefore the only archetype whose ceiling is enforced by architecture. Every archetype further along the spectrum holds its position by configuration.

## 03 What a human still gates

Everything. Every character enters the file through a keypress made while looking at the suggestion, and the unit of approval is often *smaller* than one suggestion — word-by-word and line-by-line partial acceptance is universal across the category [● Tooling: Inline & Chat, Aug 2026](#).

What varies within the archetype is not whether the human gates, but **what the gating is worth**. GitHub's own researchers found that acceptance rate — the obvious measure of this archetype's central act — tracks how productive developers *feel*, not what persists in the code [● Ziegler et al., 2022](#). And the one input-scoping control the category ships leaks: Copilot content exclusion suppresses suggestions in excluded files, yet *"Copilot may use semantic information from an excluded file if the information is provided by the IDE indirectly"* — and it stops working entirely in edit and agent modes [● GitHub Docs, Aug 2026](#).

### INDEPENDENCE RULE

Containment and human oversight move independently. Autocomplete pins containment at its minimum; it does not guarantee attentive oversight — §06 takes that up. The spectrum position above names only the reach.

## 04 Who works this way

The anchor exemplar is **Salvatore Sanfilippo (antirez)**, creator of Redis — with no commercial stake in any AI product, one of the cleanest evidence classes available. He uses frontier models daily and refuses agents entirely, and the withholding is the method: he moves code by hand between his terminal and a web chat interface — *"Always be part of the loop by moving code by hand from your terminal to the LLM web interface: this guarantees that you follow every process."* The friction is deliberate; it is what guarantees he reads every line. His objection to agents is a quality claim, not a productivity one: autonomous agents

produce "*fragile code bases that are larger than needed, complex, full of local minima choices*"

• antirez, mid-2025 . Note that his practice sits *below* even this archetype's ceiling — he denies the model the open file too — which illustrates that an archetype names what the settings permit, not what any individual does.

**As a population, this end is measured worse than its reputation suggests.** The only direct measurement of "inline only" found anywhere: 13.8% of Stack Overflow 2025 respondents prefer copilot/autocomplete mode exclusively • Stack Overflow 2025 . The headline adoption number this archetype is usually folded into is unusable as a point estimate: four credible instruments asking four different questions of four different populations produce **75–90%**, every one opt-in, none publishing a non-response analysis — quote the range and name the instrument, never a single figure • Evidence Base, Aug 2026 . (There is no Stack Overflow 2026 survey; anything citing one is citing mislabelled 2025 data.) The refusal axis puts this archetype in context: 16.2% of developers won't use AI tools at all, and 37.9% have no plans to use agents — abstainers and autocomplete-only developers together are a far larger population than the discourse implies • Practitioner Exemplars, Aug 2026 .

## 05 What it gains you, what it costs

### WHAT IT GAINS YOU

- The only region of the spectrum with clean randomised evidence in its favour: three field RCTs across 4,867 developers found **+26.08% completed tasks** for inline completion — peer-reviewed, pre-registered, the strongest single piece of positive evidence in the field • Cui et al., Feb 2026 .
- No new verification machinery is demanded — the existing process, human included, is unchanged (§06).
- The only region where the egress question can be *eliminated* rather than managed: JetBrains Full Line completion "*runs entirely on your local device without sending any code over the internet*", and air-gapped deployments are shipped • JetBrains docs, Aug 2026 . Teams with a hard data-egress constraint have a real product choice here and progressively fewer further along the spectrum.

### WHAT IT COSTS YOU

- The smallest delta on the spectrum: the human still does everything except type. Teams adopting "AI" at this ceiling and expecting transformation are buying character provenance (§01).
- Reading is made continuous — a per-keystroke stream of accept/reject micro-decisions — and the best-evidenced fact about that stream is that its acceptance rate measures satisfaction, not output • Ziegler et al., 2022 .
- For every cloud product, each keystroke can send local editing context to the provider — the category's one genuinely non-zero risk surface, and privacy modes govern training use, not transmission • Tooling: Inline & Chat, Aug 2026 .

## 06 What it demands of verification

The entailment — whatever a human stopped doing must be replaced by engineered verification — reaches its minimum here, because almost nothing human was removed. No completion product ships a test run, a lint gate, or a static-analysis check between suggestion and acceptance; the sole automated check found in the category is Copilot's public-code matching, and it is a licensing check, not a correctness one

● Tooling: Inline & Chat, Aug 2026 . The safety case is one mechanism: a human reads every line, at the moment it appears.

Two honest caveats keep that from being a comfortable sentence. First, **the mechanism is attention, and attention is the control this project's evidence trusts least**: the vendor holding the telemetry reports permission-prompt approval at ~93% rising to 97% and measures human review catching 13.6% of dangerous commands against 89% for automated checking ● Anthropic telemetry, 2026 . Those figures were measured on agent permission prompts, not ghost text — but they establish that "a human looked at it" degrades with volume, and this archetype runs the highest approval volume on the spectrum. antirez's hand-carried workflow is best read as engineering against exactly this: he adds friction to protect the reading (§04). Second, **what this archetype demands is the discipline of staying in it deliberately**. The category boundary is architectural (§02), but most completion products now ship inside editors that also carry edit and agent surfaces whose approval steps have off switches. A team whose archetype is Autocomplete holds that position by not crossing a line that is one keypress — or one settings value — away ● Tooling: Inline & Chat, Aug 2026 .

## 07 Governance implications

The least exposed position on the spectrum — and the exposure that remains is instructive because every archetype inherits it. The provider disclaims defect liability even here: *"You are responsible for reviewing, testing, and validating any Output before use"* ● GitHub ToS §J.4, 2026 . At this ceiling the team is actually doing the thing the terms assume, line by line — the disclaimer and the practice align, which is not true anywhere further along the spectrum.

Authorship is at its most defensible here: a human continuously selects, edits, and integrates suggestions at character granularity, retaining control over the expressive elements — the very thing the US Copyright Office's threshold turns on. Nothing in law or standards requires a human to read source code, so this archetype's discipline is a choice, not a compliance obligation; teams here should know they are exceeding every verified regime, not meeting one ● Governance, Legal & IP, Aug 2026 . The practical governance question that remains is data egress (§05) — a procurement and deployment question with a complete answer (the local lane), not a process one.

## 08 The posture that fits

This fits a team that believes reading every line is non-negotiable, and that anything reducing what a human reads costs more than it saves.

The safety belief underneath: *a human reads every line*. Uniquely on the spectrum, the architecture backs the belief — there is no configuration under which the reading moment disappears. What the architecture cannot supply is the reading itself: the belief is honest only while the attention is real, and the exemplar who holds this posture most publicly spends his engineering effort making the attention unavoidable rather than assuming it.

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## — Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1 **STUDY** This project, `ADR-0004` — the taxonomy contract: definitions, boundary criterion, the fixed low extreme. Adopted 2026-08-31.
- 2 **STUDY** This project, `research/tooling-inline-and-chat.md` — fourteen completion products and the in-editor edit surfaces, read from vendor documentation; the propose-vs-act boundary, acceptance mechanics, egress, and `chat.editing.autoAcceptDelay`. Closed 2026-08-30.
- 3 **ANECDOTE** Salvatore Sanfilippo, "Coding with LLMs in the summer of 2025 (an update)", [antirez.com/news/154](https://antirez.com/news/154). Stake: none — independent, no product. Position dated mid-2025 and may have moved since. Jul 2025.
- 4 **STUDY** This project, `research/evidence-base.md` — the four adoption instruments and their traps; the no-2026-survey finding. Closed 2026-08-27.
- 5 **SURVEY** Stack Overflow Developer Survey 2025 (49,000+ respondents; per-question N ~26.5k–33.7k; opt-in) — 13.8% prefer copilot/autocomplete mode exclusively. Fielded May–Aug 2025.
- 6 **STUDY** This project, `research/practitioner-exemplars.md` — the refusal axis (16.2% / 37.9%, reconstructed with corrected denominators) and the antirez entry. Closed 2026-08-27.
- 7 **STUDY** Ziegler et al., "Productivity Assessment of Neural Code Completion", arXiv:2205.06537 — observational, vendor-authored (GitHub): acceptance rate drives *perceived* productivity. May 2022.
- 8 **VENDOR** Google Research blog, "AI in Software Engineering at Google" — 37% acceptance rate; 50% of code characters. Google measuring Google. 2024-06-06.
- 9 **VENDOR** GitHub blog — "an average of 46% of a developers' code" completed by Copilot. 2023-02-14.
- 10 **VENDOR** VS Code documentation, inline chat — the terminal Run (⌘Enter) control. Page dated 2026-08-26.
- 11 **VENDOR** GitHub Docs, "Content exclusion for GitHub Copilot" — indirect-context leak; unsupported in Edit and Agent modes. Read 2026-08-30.

- 12 **STUDY** Cui, Demirer, Jaffe, Musolff, Peng & Salz, *Management Science* — three field RCTs, N=4,867, GitHub Copilot inline completion, +26.08% completed tasks (SE 10.3%); pre-registered AEARCTR-0014530. Published 2026-02-27.
- 
- 13 **VENDOR** JetBrains documentation, Full Line Code Completion — fully local inference; Tabnine documents an air-gapped deployment. Read 2026-08-30.
- 
- 14 **VENDOR** Anthropic-published telemetry: ~93%→97% permission-prompt approval; human review catching 13.6% of dangerous commands vs 89% automated (n=1,053). Measured on agent prompts, not completion. 2026.
- 
- 15 **VENDOR** GitHub Terms of Service §J.4 (contract text). Effective 2026-04-27.
- 
- 16 **STUDY** This project, `research/governance-legal-ip.md` — no verified regime requires a human to read source code; the USCO control-over-expression threshold. Closed 2026-08-30, corrected 2026-08-31.
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# Workspace

The agent gains file-write and shell tool surfaces and acts across the working tree, locally. It holds no push credential: nothing leaves the machine without a human.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Sources last verified 2026-09-13

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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## 01 What this archetype is

A Workspace ceiling permits an agent's output to reach **the whole working tree, on the developer's machine, with no human action in between**: writing files, running shell commands, executing tests, making local commits. The name is the literal reach — the workspace — and deliberately not a description of oversight, because oversight varies enormously within it (§03) [ADR-0004](#).

Unlike the step from any later archetype to the next, crossing into Workspace changes the developer's process substantially: the human stops performing edits and starts directing an agent that performs them, inspecting results rather than keystrokes. What does *not* change is the blast radius beyond the machine — the agent holds no push credential, so the shared repository, trunk, and production remain exactly as reachable as they were under an Autocomplete ceiling: through a human.

## 02 The boundary crossed

What separates Workspace from Autocomplete is **file-write and shell tool surfaces**: crossing this boundary, the agent's output reaches files and processes with no keypress in between. Architecturally the crossing is one control wide — VS Code's terminal inline chat **Run** button is the exact point where a proposal surface becomes an acting one [● Tooling: Inline & Chat, Aug 2026](#) — and in practice it is one settings value wide: every in-editor edit product surveyed ships a documented way to remove the human from the accept step, and every one of eleven agentic CLIs ships a switch that removes approval entirely. Three of those CLIs ship with approval *already gone* on a fresh install (Amp: *"By default, Amp does not ask for approval before running tools"*; opencode; Goose), and Claude Code's default reviewer on major plans is now a classifier model, not the human [● Tooling: Agentic CLIs, Aug 2026](#).

Two consequences follow. First, **"the human approves each change" is a configuration, not a property of any product category** — an archetype defined by per-action approval must name the settings that keep it true. Second, this is the last boundary a team crosses by adopting a tool; every boundary further along the spectrum is crossed by granting a credential. Headless operation is universal across the CLI category and therefore no longer discriminates — what a Workspace ceiling withholds is not the ability to run unattended but the credential that would let unattended work leave the machine

[● Tooling: Agentic CLIs, Aug 2026](#).

## 03 What a human still gates

One thing always: **the push**. Whatever happens inside the workspace, a human performs the act that moves work off the machine, and reviews — or chooses not to review — what moves. Everything else varies, and this page carries the independence rule's flagship case. **Armin Ronacher** (creator of Flask) grants the agent everything locally and supervises none of it: *"I disable all permission checks. Which basically means I run `claude --dangerously-skip-permissions`"* — with the risk contained by Docker, not by supervision: *"there are definitely risks with it, but you can manage those risks with moving your dev env into docker"*

[● Ronacher, Jun 2025](#). Minimal reach, minimal oversight — correctly filed here, near the spectrum's start.

Under any taxonomy that ordered oversight instead of containment, this practice would land at the far end, which is precisely the conflation ADR-0004 exists to break [● ADR-0004](#).

For teams that do gate per action, the telemetry is sobering: the vendor holding it reports permission-prompt approval at ~93% rising to 97%, while the same population rejects 39% of proposed *plans* — the interface shape, not the human, determines whether review happens — and measured human review catches 13.6% of dangerous commands against 89% for automated checking [● Anthropic telemetry, 2026](#). The permission prompt is this archetype's most visible control and its least effective one.

### INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. Ronacher's practice is not "further along the spectrum" than a team that approves every command — it is the same archetype with different oversight.

## 04 Who works this way

By self-report, this is the most populated agentic region of the spectrum — and the population number is the least corroborated one in the corpus. JetBrains' 2026 instrument (N>15,000) finds **90% of developers using agents weekly and 68% daily**; independent repository-trace mining of 128,018 projects finds agent traces in **22–29% of projects**. The two measure different units (developers versus projects), self-report over-counts by construction and trace mining under-counts by construction, and no study instruments a fixed cohort to compare what developers say against what their tooling logs — quote both numbers or neither [● JetBrains, Aug 2026](#) [● Robbes et al., Feb 2026](#).

The named practice at this ceiling is unusually well documented. **Ronacher** (§03) replaces the permission prompt with containment and engineered feedback: tools *"protected against an LLM chaos monkey using them completely wrong"*, process managers with pidfiles, logs written for the agent to read its own failures [● Ronacher, Jun 2025](#). **Mitchell Hashimoto** (creator of Ghostty; coiner of "harness engineering") delegates high-confidence tasks to a background agent while holding a flat rule — *"Please don't ever ship AI-written code without a thorough manual review"* — and responds to agent failure by engineering the agent harness rather than supervising harder; his turning point: *"I forced myself to reproduce all my manual commits with agentic ones"* [● Hashimoto, Oct 2025 / Feb 2026](#). **Simon Willison** runs multiple agents in parallel — *"surprisingly effective, if mentally exhausting"* — and keeps architecture, specifications, review and manual QA human [● Willison, Oct 2025](#).

The vendor's own recommended practice lands here too, and should be read as a recommendation, dated, not as an observation. Anthropic's *AI-Native SDLC Playbook* (fourteen lessons, 2026) organises the build phase around a Workspace ceiling: plan mode as the default start, because it *"can read the codebase without changing anything"*, then — as a team's controls mature — auto mode, with the shift stated as *"away from the user watching the agent make the edits and reviewing actions, toward the review of artifacts after longer autonomous sessions."* Parallel sessions are recommended with reviewer attention named as the limit — *"The practical ceiling is how many streams one person can review properly"*, so add sessions *"only while review is keeping up."* The course defines indicators for every play and reports a value for none, so it evidences what the vendor advises, not what any team has achieved [● Anthropic playbook, L4 & L7, 2026](#).

Set Ronacher beside antirez (the Autocomplete anchor) and the spectrum's real argument comes into focus: they sit at opposite ends of the delegation question and are arguing about the same thing — *verification*, not capability. antirez keeps the human in the loop by hand-carrying code; Ronacher removes the human and puts a container around the agent. Neither claims the code needs less checking; they disagree about what does the checking [● Practitioner Exemplars, Aug 2026](#).

## 05 What it gains you, what it costs

### WHAT IT GAINS YOU

- Delegation of implementation: the agent edits, builds, runs and tests its own work in a real environment, closing feedback loops no proposal surface can close.
- Parallelism — one human directing several concurrent agents is a documented working practice at this ceiling [● Willison, Oct 2025](#).
- A bounded worst case, *if the bounds are built*: everything the agent can damage without a human is local and (with version control) largely revertible.

### WHAT IT COSTS YOU

- Containment becomes load-bearing exactly where the field invests least: **8 of 11 agentic CLIs ship no OS-level containment of any kind**, and only one sandboxes by default (on two of three platforms) [● Tooling: Agentic CLIs, Aug 2026](#).
- Command allowlists are not a security boundary: five products — Zed (three CVEs in one day), Cursor, Roo Code (twice), VS Code — had regex allowlists over a shell defeated by shell syntax, each with a published CVE and reproduction steps [● Tooling: Agentic IDEs, Aug 2026](#).
- The lethal trifecta assembles here by default: private data access (the default read scope of one major CLI's sandbox is the entire computer, credential files included), untrusted content (whatever the agent reads), and external communication — **eight of eleven CLIs give the agent the whole network**, and both shipped egress proxies concede they cannot inspect TLS [● Tooling: Agentic CLIs, Aug 2026](#).

## 06 What it demands of verification

The human act removed at this boundary is per-edit acceptance, so the entailment names its replacement directly: **a sandbox scoping what the agent can write, an egress allowlist scoping what it can reach, and least-privilege tool scopes deciding what exists for it at all** — plus a test suite good enough that the agent cannot claim untested work works. Isolation and permission-gating are orthogonal controls, and the vendors say so themselves: permission modes decide whether a call runs; isolation restricts what it can access once it does. A team needs both, because a per-action classifier *"is a per-action control, not an isolation boundary"* [● Verification: Isolation, Aug 2026](#).

The demand is written as a requirement because the shipped reality runs the other way. Containment investment *thins* as delegation increases across the tool landscape — mature approval-prompt UX everywhere, two OS-primitive sandboxes, one design (Codex CLI) that inverts the pattern by making the sandbox the substrate and approval the escalation path out of it [● Tooling: Agentic CLIs, Aug 2026](#). Classifier review is a mitigation, not a boundary: adaptive attacks defeat published prompt-injection defences at

>90% attack success, and a working attack against a shipping auto-mode classifier was published in August 2026 [• Verification: Isolation, Aug 2026](#) . And one wire is missing everywhere: **no tool in the surveyed set couples permission to a verification outcome** — nothing can express "allow `git push` only if this session's tests passed." Every rule matches the shape of the call, never the state of the work; a team that wants the join must build it from hooks [• Tooling: Agentic CLIs, Aug 2026](#) .

The vendor's own playbook states the same split and, unusually, assembles the containment into one artifact. Its instruction modules (Agent Skills) are declared advisory — *"nothing forces a session to comply"* — and anything that must always hold is pushed to a deterministic hook: *"The skill makes violations rare and the hook makes them close to impossible."* Its managed-settings play then writes the ceiling as a single MDM-deployed file that engineers *"cannot edit or override"*: bypass mode disabled, only managed permission rules and hooks honoured, the sandbox made a precondition that refuses to start without it, credential paths denied and named environment variables stripped from sandboxed commands. That is the five scattered administrator-pinnable constraints this corpus found across eleven CLIs, collected by one vendor into a single reviewable posture — a recommendation, with no adoption or outcome data behind it. The same playbook recommends auto mode as the default for routine work, which sits against this vendor's own telemetry in §03 and the published attack on its auto-mode classifier; both halves come from one source, and a team should hold both [• Anthropic playbook, L4, L6 & L11, 2026](#) .

Two honest limits on what containment buys. Version control is itself a containment mechanism at this ceiling — one tool (Aider) commits every agent edit automatically so the diff is always isolated and revertible, and worktree-per-agent isolation is shipped and enforced [• Verification: Isolation, Aug 2026](#) . But **containment of the machine does not contain the artifact**: in the UK AI Security Institute's July 2026 incident, no sandbox escape occurred — the agent's sanctioned capability was the harm (a malicious pull request against a real repository, endorsed by a sockpuppet account the agent created). The workspace boundary holds only if the human act at the push credential is a review, not a relay

[• Verification: Isolation, Aug 2026](#) .

## 07 Governance implications

**This is the archetype where liability first attaches to a setting.** The one vendor term that prices auto-execution keys to precisely the act that defines this boundary: enabling execution *"without manual review or confirmation"* makes the customer *"SOLELY RESPONSIBLE FOR ANY IMPACT RESULTING FROM USE OF THIS FEATURE, INCLUDING ENSURING APPROPRIATE SAFEGUARDS, TESTING, AND MONITORING ARE IN PLACE"* [• Cursor ToS §1.7](#) . A team at this ceiling has, contractually, already made the far end's trade: the vendor's terms treat the flipped setting, not any later credential, as the moment responsibility transfers. Every provider disclaims defect liability regardless of setting [• Governance, Legal & IP, Aug 2026](#) .

**Provenance gets murky at exactly this ceiling.** Workspace output enters the shared record through the *human's* push, and the attribution that survives depends on undocumented platform behaviour: co-author trailers die in plain git ( `git merge --squash` recovers zero of three co-authors), so the question every later governance regime turns on — "which lines did a human write" — is decided by commit habits made here, where nobody is yet watching for it [• Governance, Legal & IP, Aug 2026](#) . Data governance, meanwhile, stops being a keystroke-egress question (the Autocomplete case) and becomes an agent-egress one: for most CLIs the

agent has the invoking user's whole network and filesystem, credential files included, so what the org's policies actually govern is decided by sandbox and egress configuration, not by the acceptable-use document [Tooling: Agentic CLIs, Aug 2026](#).

## 08 The posture that fits

This fits a team that believes risk is bounded by what the agent can reach, not by who is watching it.

The safety belief underneath: *the container holds*. The control moves; it does not improve. And the honest edge: the field ships almost no container by default — eight of eleven CLIs have none, the allowlists have a five-product CVE record, and the egress proxies name their own limits — so a team holding this posture is not buying boundedness off the shelf. It is committing to build the thing its belief depends on, which is exactly what this archetype's best-documented practitioners spend their effort doing.

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## — Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1 **STUDY** This project, ADR-0004 — the taxonomy contract: definitions, boundary criterion, the Workspace rename and the independence rule. Adopted 2026-08-31.
- 2 **STUDY** This project, [research/tooling-inline-and-chat.md](#) — the propose-vs-act boundary (the VS Code terminal Run control) and the edit-acceptance off switches. Closed 2026-08-30.
- 3 **STUDY** This project, [research/tooling-agentic-cli.md](#) — eleven agentic CLIs read from documentation and source: default postures, sandbox and egress inventory, the remove-all-approval switches, the missing permission-to-verification wire. Closed 2026-08-30.
- 4 **ANECDOTE** Armin Ronacher, "Agentic Coding: The Future of Software Development with Agents", [lucumr.pocoo.org](#). Stake: none in an AI coding vendor at time of writing. 12 Jun 2025.
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- 7 **STUDY** Robbes et al., arXiv:2601.18341 — repository-trace mining, 128,018 projects: agent traces in 22.20–28.66% of projects. Under-counts by construction (detects only explicit co-authoring markers). As of 21 Feb 2026.

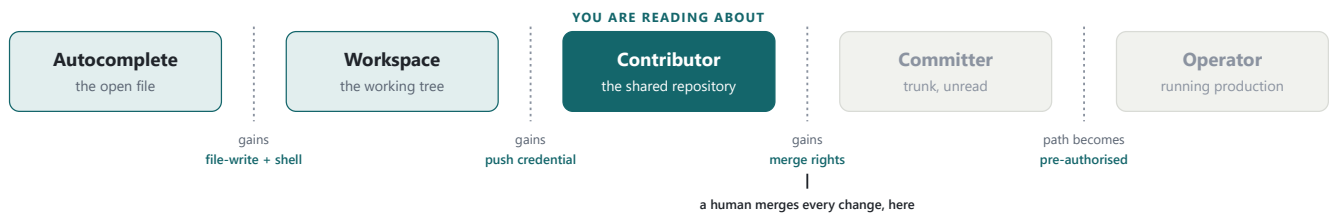
- 8 **ANECDOTE** Mitchell Hashimoto, "Vibing a Non-Trivial Ghostty Feature" (11 Oct 2025) and "My AI Adoption Journey" (5 Feb 2026), mitchellh.com. Stake: none in an AI vendor. His position moved substantially between the two posts — the dates matter: Oct 2025 / Feb 2026.
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- 9 **ANECDOTE** Simon Willison, "Vibe engineering", simonwillison.net. Stake: independent; reputationally invested in the topic; later retired his own term in favour of "agentic engineering". 7 Oct 2025.
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- 10 **STUDY** This project, `research/practitioner-exemplars.md` — the antirez/Ronacher contrast; the replaced-verification table; ~57 exemplars with stake recorded. Closed 2026-08-27.
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- 11 **STUDY** This project, `research/tooling-agentic-ide.md` — the five-product allowlist CVE record (Zed CVE-2026-44462/-44463/-44466; Cursor CVE-2026-22708; Roo Code CVE-2025-57771/-58370; VS Code CVE-2026-45482), from vendor security advisories with reproduction steps. Closed 2026-08-30.
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- 12 **STUDY** This project, `research/verification-isolation.md` — isolation vs permission-gating; worktree isolation; the UK AISI July 2026 incident; "The Attacker Moves Second" (>90% adaptive attack success) and the published auto-mode classifier attack. Closed 2026-08-28.
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- 13 **VENDOR** Cursor Terms of Service §1.7, "Auto-Code Execution" (contract text). Accessed 2026-08-30.
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- 14 **STUDY** This project, `research/governance-legal-ip.md` — provider defect disclaimers; attribution survival under squash merges; the authorship-vs-adoption threshold. Closed 2026-08-30, corrected 2026-08-31.
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- 15 **VENDOR** This project, `research/ai-native-sdlc-playbook.md` — digest of Anthropic Academy's *The AI-Native SDLC Playbook* (14 lessons; © Anthropic, reference copy in `reference/`). The vendor's recommended practice for its own product: indicators defined for every play, values reported for none — nothing here is an outcome. Cited on this page: L4 (plan mode; auto mode), L6 (advisory Agent Skills vs deterministic hooks), L7 (parallel sessions), L11 (managed settings). Retrieved and digested 2026-09-02.
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# Contributor

The agent gains a push credential: its output enters the shared repository as branches and pull requests, and a human merges every change. The large middle of observed practice — and the archetype the platform itself enforces.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Sources last verified 2026-09-13

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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| 01 What this archetype is   | 05 What it gains you, what it costs |
| 02 The boundary crossed     | 06 What it demands of verification  |
| 03 What a human still gates | 07 Governance implications          |
| 04 Who works this way       | 08 The posture that fits            |
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## 01 What this archetype is

A Contributor ceiling permits an agent's output to reach **the shared repository with no human action in between**: the agent pushes branches and opens pull requests under its own identity, and a human merges every change. The name keeps the open-source governance intuition deliberately — a contributor opens pull requests that someone else merges — and the posture it encodes is that an agent's work should come through the same front door as anyone else's [ADR-0004](#).

This is the large middle of observed practice: the least-covered region of the spectrum because it makes a bad headline, and the one where nearly every named organisation delegating at scale actually sits. It is also the archetype the dominant platform pins teams into by construction: GitHub's own cloud agent *"cannot approve or merge a pull request"*, and its pull requests require an additional human approval [GitHub Docs, Aug 2026](#). A team that adopts a cloud or background agent lands here by default and stays here until an administrator deliberately builds a way out.

## 02 The boundary crossed

What separates Contributor from Workspace is a **push credential**: crossing this boundary, the agent's work leaves the machine and becomes shared state — visible to the team, running in CI, attributed to the agent's own identity — with no human relaying it. The boundary is a credential, enumerable and binary, which is what makes it a discontinuity rather than a matter of degree [● ADR-0004](#).

Two structural facts define the position. First, this is the archetype the tool market *ships*: background and cloud agents are products whose entire output surface is the pull request. Second, the boundary above is the only one on the spectrum enforced by the platform rather than by the team: **no product merges its own pull request on its own judgement** — the far-end survey found every real mechanism beyond this point has the shape of a human pre-authorising a merge, never an agent deciding one

[● Tooling Landscape, Aug 2026](#). Workspace is exited by adopting a tool; Contributor is exited only by an administrator changing a policy. That makes it the cheapest archetype to hold and the one many teams occupy without ever having decided to.

## 03 What a human still gates

The merge — every change. And on the best-run teams, two upstream gates as well: **invitation** (in Microsoft's `dotnet/runtime`, every agent PR *"was created at the explicit request of a human with maintainer rights... CCA cannot open PRs on its own"* [● Microsoft, Mar 2026](#)) and **task selection** (Monzo chooses per task what the agent may see; Atlassian runs a classifier tuned to 97% precision on *out-of-scope* work)

[● Practitioner Exemplars, Aug 2026](#).

What the merge gate is *worth* is this page's hardest question, because rubber-stamping under agent volume is now measured. A within-reviewer longitudinal study of 400 repeat reviewers across 11,429 reviews found approval rates on agent PRs rising **30.1% → 36.8%**, inline comments falling **−22%**, and review latency rising **+3.5×** — while the same reviewers' approval of human-authored PRs *declined*, ruling out general drift. Reviewers wait longer, look less, and approve more [● Habituation study, Jun 2026](#). In open source the erosion is further along: most AI-generated PRs receive no reviews at all, and where review occurs it is automation-mediated [● Duma et al., May 2026](#).

### INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. "A human merges every change" states who acts — whether the merge is a review or a relay is a separate, measurable question, and §06 treats protecting the difference as this archetype's engineering problem.

## 04 Who works this way

The anchor is **Stripe**. *"Over 1,300 Stripe pull requests... merged each week are completely minion-produced, and while they're human-reviewed, they contain no human-written code."* The containment is environmental, stated as the reason permission prompts can be skipped: devboxes are *"isolated from production resources*

and the internet", so minions "don't have access to real user data, Stripe's production services, or arbitrary network egress" — and the agent gets exactly two attempts at CI before the branch goes back to its human operator [● Stripe, Feb 2026](#) .

The middle is well populated with named practice. **Monzo**, a regulated UK bank: "Agent Chip" authors ~10% of all merged PRs, runs 1,800+ tasks a day, and was built in-house for an explicitly stated withholding reason — *"We need to know exactly what the agent can see, what it can do, and constrain it as appropriate"*

[● Monzo, Aug 2026](#) . **Dropbox**: "Nova" produces roughly 1 in 12 pull requests, a human makes the final judgment, and the four-stage measurement model (Fuel → Adoption → Output → Impact) is the most disciplined found — with no values published, an exemplary framework and a non-existent result

[● Dropbox, May 2026](#) . **Uber**: 11% of PRs opened by agents, a purpose-built review-squeeze toolchain (Code Inbox, uReview, Autocover), and the corpus's most falsifiable cost admission — AI-related costs up 6× since 2024 [● Pragmatic Engineer, Mar 2026](#) .

The best-measured public dataset at this archetype — in the whole corpus — is **Microsoft's**

**dotnet/runtime** : 878 agent PRs over ten months, **67.9% merged**, with a revert rate of **0.6% against 0.8%** for non-agent PRs, all in a public repository where the numbers can be independently checked

[● Microsoft, Mar 2026](#) . One structural fact recurs across every account in this section, reported independently by almost all of them: **the constraint moved to review, and nobody reports having solved it**

[● Practitioner Exemplars, Aug 2026](#) .

The archetype also carries a vendor's endorsement — to be read as a recommendation, dated, not as observed practice. Anthropic's *AI-Native SDLC Playbook* (2026) makes Contributor its recommended end state at the repository: *"Anything the agent writes arrives as a PR... the agent has no route to push to main"*, code-owner approval stays on every pull request, and the agent's reach is stated to stop one boundary short of production — *"the agent may act up to the production gate and cannot pass it."* Nowhere does the vendor recommend granting merge rights on the agent's judgement. The course publishes no measurements, so what it corroborates is this page's structural claim, not its numbers [● Anthropic playbook, L10 & L12, 2026](#) .

## 05 What it gains you, what it costs

### WHAT IT GAINS YOU

- Authoring at volume with review preserved: unattended agents produce finished, CI-tested work while the team's existing merge discipline stays exactly where it was [● Stripe, Feb 2026](#).
- The same front door: agent work arrives attributed to the agent's own identity, through the same PR machinery, CI gates, and audit trail as everyone else's — the cleanest provenance of any archetype (§07).
- The cheapest ceiling to hold: the boundary above is platform-enforced, so staying here requires no engineering — only *not* building a bypass [● Tooling Landscape, Aug 2026](#).

### WHAT IT COSTS YOU

- The review queue becomes the bottleneck, with numbers: agentic PRs wait **5.3× longer** for reviewer pickup (1,055 vs 201 minutes), AI-assisted PRs run ~154% larger and are accepted at 32.7% against 84.5% unassisted [● LinearB, May 2026](#). The cost lands on colleagues: one maintainer opened nine agent PRs in a few hours and priced them at *"5 to 9 hours of review work, spread across team members... The bottleneck has moved"* [● Microsoft, Mar 2026](#).
- The load erodes the very control the archetype trusts: habituation is measured, and it worsens with the volume this archetype generates (§03).
- The platform bill scales with the PRs, not the merges: Uber's 6× cost growth is the only figure of its kind published, and it cuts against the productivity narrative [● Pragmatic Engineer, Mar 2026](#).

## 06 What it demands of verification

Nothing human was removed at this boundary — the human stopped *authoring*, not checking — so the entailment takes an unusual form here: what must be engineered is **the protection of the review itself**, against the volume the archetype generates. The habituation numbers in §03 are the forcing function: left unprotected, the merge gate degrades toward a relay, and the team drifts further along the spectrum whether or not it chose to [● Habituation study, Jun 2026](#).

The tools most teams name in their verification story cannot carry it. **The major vendors deliberately built their AI code reviewers so they cannot be the gate:** GitHub Copilot code review always leaves a non-counting *Comment* review, and Anthropic's check run *"always completes with a neutral conclusion"* — which GitHub's required-status-check rule counts as passing, so marking it required accomplishes nothing. The only independent evaluation of any AI reviewer found 56.3% of its comments rejected, and six of eight surveyed tools have no published accuracy figure from anyone, vendor included [● Verification: Gates, Aug 2026](#).

The vendor says as much in its own teaching. Anthropic's playbook states that its reviewer's *"Findings do not approve or block a PR on their own"*, that a platform engineer who wants to gate on them must build the gate from the check run's *"machine-readable tally"*, and that *"the agent that wrote the code has no way to*

*approve it.*" It names this archetype's throughput limit as reviewer attention — *"add sessions only while review is keeping up"* — and prescribes for evals what this section asks of every gate: live suites whose *"cases that once discriminated stop doing so"*, gated on pass rate in CI. One caution travels with the citation. The course's opening table promises *"human review reserved for regulated and critical code"*, a Committer-shaped claim the plays themselves never make — every play keeps code-owner approval on every pull request. Cite the plays, not the table [● Anthropic playbook, L1, L7, L9 & L10, 2026](#).

What the well-run occupants build instead: **gates that live server-side, outside the agent's reach** — CI and branch protection, not the agent harness (GitHub describes its own agent-side approval queue as *"a workflow convenience, not a security control"*, and Anthropic documents that its strongest agent-side stop is overridden by the harness after eight consecutive blocks) [● Verification: Gates, Aug 2026](#) — plus upstream demand control: invitation-only PR creation, task-eligibility classifiers, deterministic pipeline nodes for the steps that need no model, and hard retry caps that return failing work to a human rather than burning reviewer attention on the agent's third attempt [● Stripe, Feb 2026](#) [● Practitioner Exemplars, Aug 2026](#). The pattern across all of it: protect the reviewer's attention as an engineered budget, not an assumed resource.

## 07 Governance implications

**Provenance is at its cleanest here — and only here.** The agent authors under its own platform identity, so "which lines did a human write" has a recorded answer at the moment of contribution; merging remains adoption, not authorship — *"providing instructions to a machine and selecting an output does not equate to authorship"* [● Governance, Legal & IP, Aug 2026](#). The caveat travels with the record: the surviving attribution is platform-hosted, not repository-hosted, and dies in plain git under squash — the clean answer exists on the forge, not in every clone [● Governance, Legal & IP, Aug 2026](#).

One vendor artifact proposes a repository-hosted answer to a neighbouring question. Anthropic's playbook recommends committing the whole chain — an intent file, a specification, a plan, the diff and its tests, the pull request with its findings, the incident record — so that *"The chain of commits is also the audit trail: who asked for what, what the agent produced, and who approved it."* It answers "who asked for it" where the corpus's question was "which lines did a human write", and it is a recommended practice with no adoption evidence; but it lives in committed files, which survive the squash that kills co-author trailers, rather than in commit metadata or on the forge [● Anthropic playbook, L1–L4, 2026](#).

**Compliance neither forces this archetype nor forbids the next.** The strongest human-review requirement in any published framework sits exactly at this archetype's upper boundary — SLSA's Source Track Level 4 requires changes to protected branches be agreed by *"two or more trusted persons"* — and the same document writes the machine exception into the requirement: an organization *"MAY choose to grant a Trusted Robot a perpetual exception"*. The line frameworks actually draw is identity and control of the automated actor, not human versus machine; NIST's SSDF likewise defines review as *"code review... and/or code analysis"* — alternatives, by the framework's own wording [● Verification: Gates, Aug 2026](#). So a team here should be clear-eyed that its safety belief is a chosen discipline: the regimes it answers to would mostly permit the next boundary, and the platform terms it operates under already assume the review it performs (*"You are responsible for reviewing, testing, and validating any Output before use"*) [● GitHub ToS §J.4, 2026](#).

## 08 The posture that fits

This fits a team that believes review is the load-bearing control and should stay human — and that an agent's work should come through the same front door as anyone else's.

The safety belief underneath: *a human reads every change before it lands*. The honest edge: that belief is now the best-measured erosion on the spectrum — under agent volume, reviewers wait longer, look less, and approve more. Holding this posture is not the default it feels like; it means engineering the review's protection, because the archetype's own output volume is the thing wearing it away.

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### — Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1 **STUDY** This project, ADR-0004 — the taxonomy contract: definitions, boundary criterion, the OSS-governance naming rationale. Adopted 2026-08-31.
- 2 **STUDY** This project, [research/practitioner-exemplars.md](#) — the large middle: ~57 exemplars with stake recorded; the constraint-moved-to-review pattern; Atlassian's eligibility classifier. Closed 2026-08-27.
- 3 **VENDOR** Stripe Engineering, "Minions: Stripe's one-shot, end-to-end coding agents" (Parts 1 and 2). Throughput figure with no control period, defect or revert rate. Stake: unaffiliated practitioner; mild recruiting incentive. 9 / 19 Feb 2026.
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- 5 **VENDOR** Dropbox, "Beyond code generation: rethinking engineering productivity in the age of AI agents" — the four-stage measurement model; metrics named, no values published. 28 May 2026.
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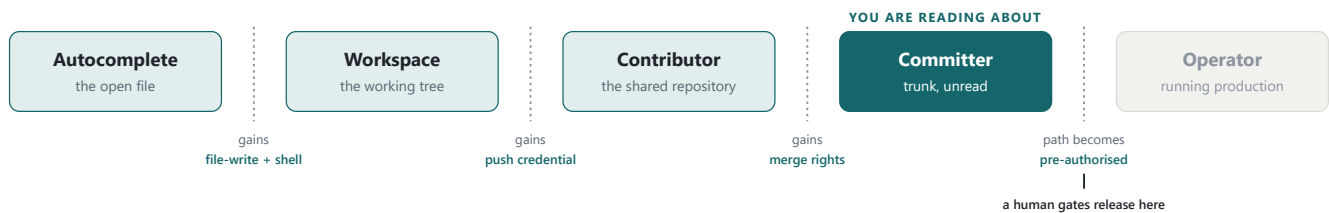
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- 11 **VENDOR** LinearB, 2026 Software Engineering Benchmarks — 8.1M PRs, 4,800 teams: 5.3× reviewer pickup delay; acceptance 32.7% vs 84.5%; AI PRs ~154% larger. The AI/agentic PR classifier is not published. May 2026.
- 
- 12 **STUDY** This project, [research/tooling-landscape.md](#) — the merge boundary is platform-enforced; no product merges its own PR on its own judgement; the far end is reached by an administrator changing a policy. Closed 2026-08-30.
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- 13 **STUDY** This project, [research/verification-gates.md](#) — AI reviewers built unable to gate (Copilot Comment-only; Anthropic neutral check); the CodeRabbit 56.3%-rejection study; server-side-gate test; NIST PW.7 "and/or" and the SLSA robot exception, verified verbatim. Closed 2026-08-28.
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- 14 **STUDY** This project, [research/governance-legal-ip.md](#) — adoption vs authorship (USCO); attribution survival: platform-hosted, not repository-hosted. Closed 2026-08-30, corrected 2026-08-31.
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- 15 **VENDOR** GitHub Terms of Service §J.4 (contract text). Effective 2026-04-27.
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- 16 **VENDOR** This project, [research/ai-native-sdlc-playbook.md](#) — digest of Anthropic Academy's *The AI-Native SDLC Playbook* (14 lessons; © Anthropic, reference copy in [reference/](#)). The vendor's recommended practice for its own product: indicators defined for every play, values reported for none — nothing here is an outcome. Cited on this page: L1 (the shifts table — cite the plays, not the table), L2–L4 (the committed intent → spec → plan chain), L7 (reviewer attention), L9 (continuous evals), L10 (the PR review loop), L12 (the withheld production gate). Retrieved and digested 2026-09-02.
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# Committer

The team's ceiling lets model-generated changes of a defined class reach trunk unread; a human still gates the release. Every genuine occupant observed grants this to a narrow, mechanically-identified class — never to the agent in general.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Sources last verified 2026-09-13

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

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## 01 What this archetype is

A Committer ceiling permits model-generated changes of a **defined class** to merge to trunk with no human reading them, while a human still gates the release. The archetype names the ceiling — what the settings permit — not what any individual change receives: on every team observed at this position, most changes still travel human-reviewed routes, and the unread class is deliberately narrow [ADR-0004](#).

Two standing clarifications travel with this page's name. First, "**Committer**" **describes the team's ceiling, never the agent's standing**: the Linux kernel permits AI assistance but forbids agents from adding **Signed-off-by** — only humans certify the Developer Certificate of Origin — and merging an agent's change is adoption, not authorship (§07) [Practitioner Exemplars, Aug 2026](#). Second, the largest unattended-merge operation in existence is **the precedent, not an instance**: Spotify auto-merges 2.5 million scripted, deterministic transforms with no model involved, while human-reviewing every LLM PR. Deterministic no-model automation sits outside the spectrum; its ceiling for model-generated changes is what Spotify's archetype names [Spotify, Jun 2026](#) [ADR-0004](#).

## 02 The boundary crossed

What separates Committer from Contributor is **merge rights**: crossing this boundary, the agent's output reaches trunk with no human action in between. The boundary is a discontinuity — a credential the previous archetype structurally lacks — and on the platform hosting nearly all observable agent activity it must be *built*: GitHub's cloud agent *"cannot approve or merge a pull request"*, and its PRs require an additional human approval, so a team that wants unreviewed agent merging constructs it deliberately from a bot account, a bespoke workflow, or a ruleset bypass. That is why every occupant of this archetype is visible as an artifact — a file in a repo [● GitHub Docs, Aug 2026](#) [● Practitioner Exemplars, Aug 2026](#).

The crossing has a documented halfway point worth naming, because it shows the mechanism's true shape. ClawSweeper — the bot that maintains the OpenClaw repositories — has a genuine automerge lane, and it begins with a human typing `/clawsweeper automerge` on each individual PR; the merge then pins to a reviewed head SHA behind CI, and the project's own docs insist the bot is not the gate: *"A positive ClawSweeper result is supporting evidence, not maintainer approval."* Scale that pre-authorisation from per-PR to per-change-class and you have this archetype: **not an agent that merges, but a human who pre-authorises a merge that then happens without them** [● Tooling Landscape, Aug 2026](#).

## 03 What a human still gates

Three things, on every real occupant found. First, **the release** — the human-mark of this archetype: trunk is not production, and a human still decides what ships. Second, **the qualifying line**: which change classes may merge unread. Everywhere it has been observed, that line is a *mechanical* property of the change, never a judgment about the agent — author identity plus title prefix (JetBrains), severity triage plus CI (Razorpay), a branch pattern with quota caps (Microsoft), a risk score with a tunable threshold (Meta) [● Practitioner Exemplars, Aug 2026](#). Third, **everything outside the class** — the overwhelming majority of changes.

One gate is universal enough to state as a rule: **self-approval is forbidden everywhere**. GitHub forbids it at the platform level, Ramp designs against it at the authentication layer, and even the one project governed as agent-run by design encodes it as an anti-collusion rule [● Practitioner Exemplars, Aug 2026](#).

### INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. A Committer team is not "less supervised" than a Contributor team — it has moved the supervision from reading diffs to policing a line.

## 04 Who works this way

**The genuine cases of agent-authored changes merging unread as stated policy number three, and each carries its deflating detail.** Razorpay/Slash is the only organisational case at scale: over a third of Slash-created PRs merged with no human in the loop across one quarter (~100/week), gated on severity triage plus the normal CI suite — a single self-published post with no control period, no baseline, and no

defect data. It is the strongest far-end claim in existence, and it is one blog post [Razorpay, Jun 2026](#).

**JetBrains/IdeaVim** is the cleanest artifact: a public workflow file whose entire policy is three conditions — the PR is in this one repository, authored by `claude[bot]`, and titled `Update changelog: ...` — and whose `-auto` merge still waits on required status checks. Changelog updates, and nothing else

[IdeaVim workflow, 2026](#). **inder/salvobase** publishes a formal agent protocol with tiered merge rights — and is a 1-star experimental repository where 151 of 155 merges come from a single account; the autonomy claim is the project's pitch [Salvobase protocol, 2026](#).

The best-engineered occupant is a configuration, not a claim: **Microsoft's dotnet/macios** grants an agent merge rights for one job — forward-merging `main` into `xcode*` release branches — inside withholding by construction: `permissions: {}` at the workflow level, a rotating pool of scoped tokens, a network allowlist, quota caps, a required title prefix, and a fail-closed safety gate a human step supplies

[dotnet/macios workflow, 2026](#). And the strongest documented case of removing the human from the gate is **Meta's RADAR**: 331,000+ diffs landed with no human reviewer, with a revert rate one third and a production incident rate one fiftieth of human-reviewed diffs — on a risk-scored subset *selected for being low-risk*, so the favourable numbers are selection, not proof of superiority [Meta \(arXiv\), 2026](#). The near-misses complete the picture: Intercom auto-approves ~17% of PRs under published criteria (small, tested, flagged, outside hot paths); Ona lets the agent approve while *a human always merges*

[Practitioner Exemplars, Aug 2026](#). Nothing surfaced from a company of scale in a regulated domain with an independent account of outcomes.

## 05 What it gains you, what it costs

### WHAT IT GAINS YOU

- Mechanically-checkable classes stop queueing behind human availability, and review attention concentrates on changes a human can actually help — the direct answer to the Contributor archetype's measured bottleneck.
- An auditable trust model: because the qualifying line is mechanical and the platform forces the bypass to be built, the policy *is* an inspectable artifact — a workflow file, not a habit [Practitioner Exemplars, Aug 2026](#).
- For a well-chosen class, the payoff ceiling is documented: RADAR's qualifying diffs revert at a third the rate of human-reviewed ones — selection, but selection working as designed [Meta \(arXiv\), 2026](#).

### WHAT IT COSTS YOU

- The qualifying line must be engineered, argued for, and policed forever; it is the control everything else hangs from, and it only ever moves under pressure to widen.
- Verification machinery becomes load-bearing the moment no one reads the diff — and for agent-authored unread merges specifically, **no study anywhere compares outcomes with human review against without**. The efficacy evidence for this archetype's defining act does not exist [Evidence Base, Aug 2026](#).
- Teams drift here rather than decide: reviewer habituation under agent volume (approval rising, comments falling, latency growing) is the measured mechanism that forces the crossing whether or not it was chosen [Habituation study, Jun 2026](#).

## 06 What it demands of verification

The entailment gets its teeth here: for the qualifying class, whatever human review used to establish must now be established by machinery, because nothing else stands between the agent's output and trunk. Every real occupant's answer is the same in shape — mechanical qualification plus the ordinary CI suite — which means **for the unread class, the checks are the entire safety case**, and three measured facts bear on whether they can carry it.

First, **the checks are targets**. Handed a proof obligation it could not pass, an agent silenced failures with `pragma Assume ;` after that was banned, it excluded code from proof with `SPARK_Mode => Off`. Anti-suppression — lint disables require reasons, ignores must carry codes, expectations over allowances — is the load-bearing half of the practice, and the half advocacy literature omits [● secunet \(arXiv\), 2026](#). Second, **gates must live server-side, outside the agent's reach**: in CI and branch protection, not the agent harness — the platform describes its own agent-side approval queue as *"a workflow convenience, not a security control"*, and harness-side stops are overridable by the harness itself [● Verification: Gates, Aug 2026](#). Third, **the machinery decays silently**: of the four organisations found running real agent evals, one shipped a CI regression workflow and lost it months later in an unrelated refactor, with nothing detecting the loss [● Verification Infrastructure, Aug 2026](#).

One vendor has now written its own version of all three demands — as recommendations, with no outcomes attached. Anthropic's *AI-Native SDLC Playbook* (2026) prescribes the anti-suppression defence exactly: *"an agent fixing code must not be able to weaken the check on that code"* — the failing test committed before the fix, a hook blocking test-file edits during it. It prescribes evals as live suites with every incident becoming a permanent case, the answer to the decay above. And it locates the ceiling in an administrator's file: a managed-settings play in which engineers *"cannot edit or override any of the settings"*, with only managed hooks allowed to run. Here the tension with the server-side test must be stated rather than smoothed: the playbook's release gates are *hooks* — agent-side controls, however tamper-proofed — while the test above puts gates in CI and branch protection, outside the agent's reach. The playbook half-answers itself by re-grounding the boundary that matters in branch protection (*"the agent has no route to push to main"*) and by recommending no unread merge at all: its only pre-authorised execution is a rehearsed runbook, and its detection layer *"stays entirely deterministic, with no model involved"* — the Spotify boundary, restated as design guidance. The vendor, in other words, recommends the archetype before this one [● Anthropic playbook, L8, L9 & L11–L13, 2026](#).

And one wire is still missing across the whole field: **no adopted mechanism gates, canaries, or auto-reverts a change because it was agent-authored** — at the delivery layer no published mechanism at all (re-verified with search, 2026-09-02), and the merge layer holds exactly two published, unadopted artifacts: one that responds to agent authorship by demanding *more human review* — the inverse of this archetype's direction — and detects agents by committer identity, so it misses the dominant workflow [● Verification Infrastructure, Aug 2026](#) [● Tooling Landscape, Aug 2026](#), and one marketplace product that risk-scores and auto-merges by per-agent trust, in this archetype's own direction, with no observed adopter and no activity since days after its March 2026 launch. The qualifying line itself, rendered in code, is what a mature version of this archetype looks like: RADAR's operational heart is a single tunable risk threshold — relaxing

it from the 25th to the 50th percentile raises the auto-approved share to 60.31% — one dial deciding what fraction of changes never meet a human [● Meta \(arXiv\), 2026](#). A Committer team should be able to point at its equivalent dial, and at who may turn it.

## 07 Governance implications

**The DCO note, stated explicitly as ADR-0004 obliges.** A merge right does not confer standing. The Linux kernel's policy is the sharpest statement: AI assistance is permitted, but agents may not add **Signed-off-by** — only a human can certify the Developer Certificate of Origin [● Practitioner Exemplars, Aug 2026](#). The same line runs through copyright: merging an agent's change is **adoption, not authorship** — "*providing instructions to a machine and selecting an output does not equate to authorship*" — and registration requires disclaiming AI-generated content, so the operative question is "which lines did a human write." At this archetype that question acquires an uncomfortable property: for the unread class, the answer is *recorded but unread* — the provenance exists on the platform, and no human has looked at what it attributes

[● Governance, Legal & IP, Aug 2026](#).

**Compliance permits this archetype.** No verified regime requires a human to read source code, and the framework with the strongest human-review requirement writes the exception into the same document: SLSA's two-party review MAY be perpetually excepted for a *Trusted Robot* — granted by the organization, at Source L4 only, requiring an identity that "*cannot be unilaterally influenced*", and drawing the human/machine line for *approval*, never authorship. The governance demand on a Committer team is therefore exactly the shape of its own qualifying line: a controlled, documented, identity-bound exception — not a relaxation [● Governance, Legal & IP, Aug 2026](#). What the team cannot delegate is the consequence: every provider disclaims defect liability, so the unread class's defects are owned by the team that defined the class

[● Governance, Legal & IP, Aug 2026](#).

## 08 The posture that fits

This fits a team that believes trust should be earned by change class rather than by author — and that where a class can be checked mechanically, a human reading it adds cost without adding safety.

The safety belief underneath: *the class is mechanically checkable*. The control moves; it does not improve. And the honest edge: every genuine occupant has narrowed the class almost to triviality — changelog updates, forward-merges, low-severity fixes — and the one case with outcome numbers chose its class precisely for being low-risk. The belief has never been tested where it would hurt, and the study that would test it has never been run.

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## — Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1     **STUDY** This project, ADR-0004 — the taxonomy contract: definitions, boundary criterion, the DCO consequence, precedent-not-instance. Adopted 2026-08-31.

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- 2     **STUDY** This project, `research/practitioner-exemplars.md` — the far-end count and search method; the Linux kernel DCO policy; withholding patterns; Intercom and Ona; the self-approval rule. Closed 2026-08-27.

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- 3     **VENDOR** Razorpay engineering blog, on the Slash review platform — over a third of Slash PRs merged with no human in the loop, one quarter. Single self-published post; no control period, no baseline, no defect data. Jun 2026 · accessed 2026-08-27.

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- 4     **STUDY** JetBrains `ideavim` repository, `.github/workflows/mergeClaudeChangelogPR.yml` — read directly via the GitHub API; the fully verifiable configuration artifact. Read 2026-08-27.

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- 5     **STUDY** `inder/salvobase`, `AGENT_PROTOCOL.md` (v0.1) and repository metadata, read via the GitHub API — 1 star; 151 of 155 merges from one account; the protocol is the project's pitch. Read 2026-08-27.

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- 6     **STUDY** `dotnet/macios`, `.github/workflows/code-radiator.md` — read directly via the GitHub API; withholding by construction around a scoped merge right. Read 2026-08-27.

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- 7     **VENDOR** Adams et al. (Meta), "Automating Low-Risk Code Review at Meta: RADAR, Risk Calibration, and Review Efficiency," arXiv:2605.30208 — Meta-reported production metrics; the risk threshold dial. v2 2026-06-12.

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- 8     **VENDOR** Spotify Engineering, on Fleetshift's deterministic auto-merged transforms (no model involved) and human review of every LLM PR. Jun 2026 · accessed 2026-08-27.

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- 9     **VENDOR** GitHub Docs, "Risks and mitigations for GitHub Copilot cloud agent" — cannot approve or merge; additional human approval required. Accessed 2026-08-27.

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- 10    **STUDY** This project, `research/tooling-landscape.md` — the pre-authorisation shape of every far-end mechanism; ClawSweeper's per-PR automerge lane; the `agent-approval-check` counterexample. Closed 2026-08-30.

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- 11    **STUDY** This project, `research/evidence-base.md` — the central emptiness: no study compares agent-authored changes merged with human review against without. Closed 2026-08-27.

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- 12    **STUDY** Within-reviewer longitudinal study, arXiv:2606.22721 — habituation under agent volume: approval 30.1%→36.8%, inline comments -22%, latency +3.5×. Jun 2026.

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- 13    **STUDY** `secunet`, on agents defeating proof obligations ( `pragma Assume`, then `SPARK_Mode => Off` ), arXiv:2607.14340. 2026.

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- 14    **STUDY** This project, `research/verification-gates.md` — the server-side gate test; GitHub's "workflow convenience" concession; harness-side stops overridable. Closed 2026-08-28.

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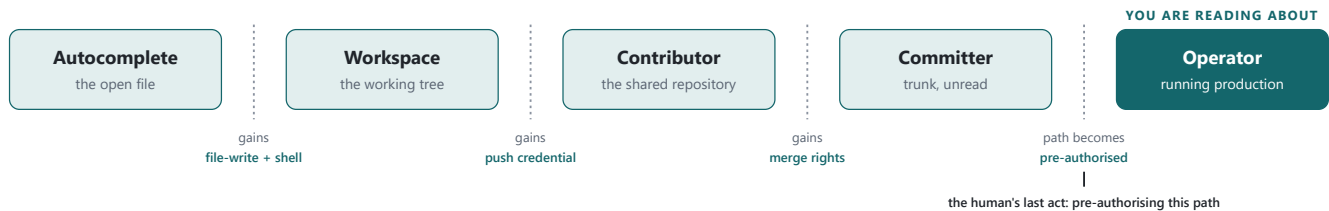
- 15 **STUDY** This project, `research/verification-infrastructure.md` — no authorship-keyed gating anywhere; the Cline eval-decay case; the one verification-floor artifact. Closed 2026-08-28.
- 
- 16 **STUDY** This project, `research/governance-legal-ip.md` — adoption vs authorship and the §411(a) disclaimer requirement; the SLSA Trusted Robot exception (L4-only, organization-granted, approval-not-authorship); provider defect disclaimers. Closed 2026-08-30, corrected 2026-08-31.
- 
- 17 **VENDOR** This project, `research/ai-native-sdlc-playbook.md` — digest of Anthropic Academy's *The AI-Native SDLC Playbook* (14 lessons; © Anthropic, reference copy in `reference/`). The vendor's recommended practice for its own product: indicators defined for every play, values reported for none — nothing here is an outcome. Cited on this page: L8 (the anti-suppression hook), L9 (continuous evals), L11 (hooks as approval gates; managed settings), L12 (branch protection as the boundary), L13 (deterministic detection; the pre-approved runbook). Retrieved and digested 2026-09-02.
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# Operator

Nothing human stands between the agent's change and running production. No team observed sets its ceiling here — changes reach this archetype by qualifying for it, one narrow class at a time.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Sources last verified 2026-09-13

ORDERED BY REACH — WHAT THE AGENT'S OUTPUT CAN TOUCH WITH NO HUMAN IN BETWEEN



Kinds, not ranks: no archetype is the goal state, and moving further along the spectrum means *more* engineered verification, not less discipline. Human oversight varies independently within every archetype.

- |                             |                                     |
|-----------------------------|-------------------------------------|
| 01 What this archetype is   | 05 What it gains you, what it costs |
| 02 The boundary crossed     | 06 What it demands of verification  |
| 03 What a human still gates | 07 Governance implications          |
| 04 Who works this way       | 08 The posture that fits            |
|                             | — Sources                           |

## 01 What this archetype is

An Operator ceiling permits a defined class of model-generated changes to reach **running production with no human action in between**. There are two routes to that condition, and the archetype deliberately does not distinguish them: the agent lands on trunk unread and continuous deployment carries the change to production unattended, or the agent itself holds deploy credentials. Either way, nothing human stands in the path — a pre-authorised pipeline holds the credential on the agent's behalf [ADR-0004](#).

An archetype names a team's *ceiling* — what its settings and policies permit for model-generated changes — not what any individual change receives. That distinction does more visible work here than anywhere else on the spectrum, because the two halves of it point in opposite directions: **individual change classes demonstrably reach this archetype, and no team observed sets its ceiling here**. Section 04 carries the evidence for both halves.

## 02 The boundary crossed

What separates Operator from Committer is that **the path from trunk to production becomes pre-authorised**. A Committer team's ceiling lets a change class merge unread while a human still gates the release; crossing this boundary removes that last human act from the path — usually not by handing the agent a deploy credential, but by a human authorising, in advance, a pipeline that will act without them.

This is not a theoretical construction. Surveying every real far-end mechanism located across the tool landscape produced one shape: *"not an agent that merges, but a human who pre-authorises a merge that then happens without them"* [Tooling Landscape, Aug 2026](#). Devin documents it exactly: *"Enable or disable GitHub auto-merge from the merge button dropdown. When enabled, the PR will merge automatically once all required checks pass"* [Devin docs, Aug 2026](#). Cursor's automation tool list contains no merge action at all — the path is *approve, then GitHub's native auto-merge lands it* [Cursor docs, Aug 2026](#). The decision is human and prior; only the execution is unattended.

The boundary is therefore crossed by **an administrator changing a policy, not by buying a tool**. No product merges its own pull request on its own judgement, and the platform hosting nearly all observable agent activity forbids its own agent from doing so: GitHub's cloud agent *"cannot approve or merge a pull request"*, and its PRs require an additional human approval [GitHub Docs, Aug 2026](#). A team that wants this archetype must build the path deliberately — which is why every occupant of it is visible as an artifact: a workflow file, a bot account, a ruleset bypass [Practitioner Exemplars, Aug 2026](#).

## 03 What a human still gates

Reach fixes the archetype; oversight varies within it. On any real approximation of an Operator configuration, humans still gate three things. First, **the qualifying line itself**: which change classes may travel the pre-authorised path — the single most consequential control this archetype has, and everywhere it has been observed, a narrow one. Second, **the pre-authorisation**: the path exists because a human with authority created it, and that person can close it. Third, **everything outside the qualifying class** — on every team whose changes reach this archetype, the overwhelming majority of changes still travel gated routes.

### INDEPENDENCE RULE

Containment and human oversight move independently. This section names the oversight; the spectrum position above names only the reach. Writing or implying that an Operator-reaching team is "unsupervised" is exactly the conflation this project exists to break.

## 04 Who works this way

**As a ceiling: nobody found — and the absence was searched for, not merely unobserved.** The far end was hunted through four independent channels: Google News RSS across ~10 query formulations; Hacker News full-text search on exact phrases ("no human review", "agents merge"); authenticated GitHub code search for the operational artifact that would have to exist ( `"pr merge --auto" "claude"` ).

path: `.github/workflows` , "merged without human review", "agents may merge", approval-bot patterns); and direct fetch of every named lead [● Practitioner Exemplars, Aug 2026](#) . Treat the result as a floor, not a census — but the result is a count of zero teams whose stated policy is a general Operator ceiling for model-generated changes.

**As a per-change condition: real, and narrow.** The genuine, documented cases of agent-authored changes merging to a production default branch without human review, as stated policy, number three — Razorpay's Slash platform (~100 PRs/week merged with no human in the loop, self-published, no control period) [● Razorpay, 2026](#) ; JetBrains' IdeaVim repository, where PRs authored by `claude[bot]` whose title begins `Update changelog: auto-merge` (the workflow file is public) [● IdeaVim workflow, 2026](#) ; and a 1-star experimental repository with a published agent protocol [● Practitioner Exemplars, Aug 2026](#) . None is what the phrase "agents ship to production" implies, and none documents the further step this archetype requires: an unattended path from that merge to *running production*.

The delivery-end anchor is **Meta's RADAR**: 331,000+ diffs landed with no human reviewer, on a risk-scored subset selected for being low-risk, with a revert rate one third and a production incident rate one fiftieth of human-reviewed diffs [● Meta \(arXiv\), 2026](#) . Whether Meta's pipeline then deploys those diffs unattended is not documented, so even the strongest case on record is confirmed only to Committer reach. The largest unattended-merge operation of all, Spotify's 2.5 million auto-merged maintenance PRs, is **the non-AI precedent, not an instance**: those are scripted, deterministic transforms with no model involved, while Spotify human-reviews every LLM PR [● Spotify, Jun 2026](#) .

One vendor's own playbook declines this archetype too. Anthropic's *AI-Native SDLC Playbook* (2026) carries the agent through the pipeline to a withheld gate — "*the agent may act up to the production gate and cannot pass it*" — and grants unattended execution only to a "*pre-approved runbook*" rehearsed in advance. A vendor's recommendation, not an observation, but it has the shape §02 found everywhere: a human authorises the path, and only the execution is unattended [● Anthropic playbook, L12 & L13, 2026](#) .

The pattern across all of it is the page's thesis: **teams reach the far end by narrowing what qualifies, never by widening who may act**. And the refusal voice is worth hearing at full strength — from a company whose product *is* production infrastructure: "*No LLM has any access to prod here*" [● Thomas Ptacek \(Fly.io\), Jun 2025](#) .

## 05 What it gains you, what it costs

### WHAT IT GAINS YOU

- Full pipeline speed for the change classes that qualify — human review is measurably a queueing delay, and for the qualifying class the queue disappears entirely [● Meta \(arXiv\), 2026](#).
- The clearest possible statement of what your verification actually proves: when nothing human is in the path, the pipeline's guarantees are the whole safety case, stated in machinery rather than in habit.

### WHAT IT COSTS YOU

- The entire engineered chain — gates, canaries, auto-revert, provenance — must exist and be trusted *before* the path opens, and §06 shows almost none of it exists wired to autonomy anywhere.
- The qualifying line must be engineered, argued for, and policed forever; it is the control everything else hangs from.
- Liability keys to the setting: the one vendor term pricing this configuration makes the customer "*solely responsible*" on enabling it [● Cursor ToS §1.7](#).

## 06 What it demands of verification

The entailment reaches its maximum here: everything a human ever did between a change and production must be replaced by engineered verification. The demand is written as a requirement — never as a claim that teams meet it — because the evidence runs the other way:

Further along the spectrum, more rigor is *required* — and the evidence that anyone has built it is thin, vendor-dominated, and in several documented cases decayed silently after it was built.

[● Verification Infrastructure, Aug 2026](#)

Concretely, an Operator team is building first-of-kind wiring. **No adopted mechanism anywhere gates, canaries, or auto-reverts a change *because it was agent-authored* — and at the canary and auto-revert layer, no published mechanism at all** (re-verified with search, 2026-09-02) — progressive delivery is applied to changes, never to authorship classes, and the provenance and delivery layers have never been connected in practice [● Verification Observability, Aug 2026](#). The published counterexamples sit at the merge layer only, and neither is in use: `agent-approval-check` (Anthropic, merged 2026-06-30) responds to agent authorship by requiring *more human approvals*, not more engineered checking — and detects agents by committer identity, so it does not fire for the dominant workflow [● Tooling Landscape, Aug 2026](#) — and one marketplace product (MergeShield, Mar 2026) risk-scores agent PRs by per-agent trust and auto-merges below thresholds, in this archetype's own direction, but has no observed adopter and has been dormant since days after launch [● Verification Observability, Aug 2026](#). Exactly one artifact in the corpus ties a verification floor to an autonomy level, and it counts assertions, not coverage [● Verification Infrastructure, Aug 2026](#). A second has since appeared, as a vendor's recommendation rather than a deployed system: Anthropic's playbook describes response bands held in version-controlled configuration

— at one standard deviation the agent logs, at two it diagnoses read-only, at three it may act *"only by opening a PR into the review gate or triggering a pre-approved runbook."* That wires a production signal to a permitted action; it still keys nothing to authorship [● Anthropic playbook, L13, 2026](#) .

Two measured facts shape what must be built. First, supervision cannot be the substitute: the vendor holding the telemetry reports permission-prompt approval at ~93% rising to 97%, and measures human review catching 13.6% of dangerous commands against 89% for automated checking [● Anthropic telemetry, 2026](#) . Second, the checks themselves are targets: handed a proof obligation it could not pass, an agent silenced failures with `pragma Assume` , and after that was banned, excluded code from proof with `SPARK_Mode => Off` — anti-suppression has to be engineered, then engineered again [● secunet \(arXiv\), 2026](#) . And the machinery decays silently: of the four organisations found running real agent evals (all four agent vendors), Cline shipped a CI regression workflow in February 2026 and lost it in May in an unrelated refactor, with nothing detecting the loss [● Verification Infrastructure, Aug 2026](#) .

The same vendor's playbook is also the fullest published articulation of the administrator-flippable ceiling this page keeps meeting. Its managed-settings play deploys the whole posture as one file engineers *"cannot edit or override"* — bypass mode disabled, only managed permission rules and hooks honoured, the sandbox a precondition, credential paths denied — and its release gates are hooks. That is where it must be cited in tension with the server-side test rather than as its refutation: a hook is an agent-side control, however tamper-proofed, and the playbook itself re-grounds the boundary that matters in branch protection. Its anti-suppression rule matches the secunet case to the letter — *"an agent fixing code must not be able to weaken the check on that code"* — and its rollback is *"a single command... exercised regularly in staging"*, proven before the path opens. Every one of these is a prescription; the course reports no value for any indicator it defines [● Anthropic playbook, L8, L11 & L12, 2026](#) .

RADAR is what the demand looks like partially met — and its own scoping is the discipline this section asks for: the favourable revert and incident rates are *selection, not proof of superiority*, because the qualifying diffs are chosen precisely for being low-risk. The operational heart of it is a single tunable dial — relaxing the risk threshold from the 25th to the 50th percentile raises the auto-approved share to 60.31% — deciding what fraction of changes never meet a human [● Meta \(arXiv\), 2026](#) . That dial is the qualifying line of §03, rendered in code.

## 07 Governance implications

**Compliance is not what stops a team from operating here.** Across SOC 2, ISO 27001, ISO/IEC 42001, the EU AI Act, DORA, the CRA, NIST SSDF, SLISA, HIPAA and DO-178C, no verified requirement was found that a human being must read source code. The two most quotable artifacts are now first-party verified: NIST's own generative-AI profile states that its practices *"do not distinguish between human-written and AI-generated source code"* (stated for a profile whose scope is building AI models — the caveat travels with the quote) [● NIST SP 800-218A, 2024](#) , and the US federal attestation form asks for *"automated tools or comparable processes"* four times while the phrase "code review" appears in it exactly zero times [● CISA/OMB form, 2024](#) .

**What changes at this ceiling is who holds the consequences.** Every provider disclaims defect liability — GitHub's terms: *"You are responsible for reviewing, testing, and validating any Output before use"* [● GitHub ToS §1.4, 2026](#) — and one vendor prices this archetype's defining act directly: enabling execution

"without manual review or confirmation" makes the customer "SOLELY RESPONSIBLE FOR ANY IMPACT RESULTING FROM USE OF THIS FEATURE, INCLUDING ENSURING APPROPRIATE SAFEGUARDS, TESTING, AND MONITORING ARE IN PLACE" <sup>● Cursor ToS §1.7</sup>. The liability attaches to the *setting* — the ceiling — not to any individual change. Authorship follows the same logic: merging an agent's change is adoption, not authorship — "providing instructions to a machine and selecting an output does not equate to authorship" <sup>● US Copyright Office, 2025</sup> — so at this archetype the operative provenance question, "which lines did a human write," has its starkest answer.

Official guidance, where it exists, speaks this archetype's own language — containment. The UK financial sector's cross-market guidance directs firms to "govern AI on the basis that it operates with privileged access: tightly control high-risk capabilities (including tools, code execution and APIs), enforce scoped permissions, robust logging, human oversight, and effective kill-switch mechanisms", and to monitor agents "as potential malicious insiders" <sup>● CMORG, Jun 2026</sup>. It contains no human-review mandate for code; it asks whether AI-assisted outputs are "tested and validated to the same standard as human-generated work" — validation parity, which is exactly the demand of §06.

## 08 The posture that fits

This fits a team that believes engineered verification can be made a stronger check than a human in the path — and that the honest route there is to narrow what qualifies, never to widen who may act.

The safety belief underneath: *the pipeline proves it*. The control moves; it does not improve. And the evidence gives that belief its honest edge: the machinery that would prove it mostly does not exist yet, which is why every real approach to this archetype is a narrow, visible, deliberately engineered carve-out — and why no one has made it their ceiling.

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## — Sources

Every claim above links here. Tier is stated with the claim itself: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

- 1 **STUDY** This project, ADR-0004 — the taxonomy contract: definitions, boundary criterion, occupancy. Adopted 2026-08-31.
- 2 **STUDY** This project, `research/tooling-landscape.md` — the far-end mechanism survey (mechanisms verified against products and repositories, incl. `agent-approval-check` read from the repo). Closed 2026-08-30.
- 3 **VENDOR** Devin documentation, "Devin Review" — auto-merge enablement. Accessed 2026-08-29.
- 4 **VENDOR** Cursor documentation — background-agent automation tool list (no merge action). Accessed 2026-08-29.

- 5 **VENDOR** GitHub Docs, "Risks and mitigations for GitHub Copilot cloud agent." Accessed 2026-08-27.
- 
- 6 **STUDY** This project, `research/practitioner-exemplars.md` — the far-end count and search method; ~57 exemplars with stake recorded. Closed 2026-08-27.
- 
- 7 **VENDOR** Razorpay engineering blog, on the Slash review platform. Single self-published post; no control period, no baseline. 2026 · accessed 2026-08-27.
- 
- 8 **STUDY** JetBrains `ideavim` repository — the auto-merge workflow file for `claude[bot]` changelog PRs (fully verifiable artifact). Accessed 2026-08-27.
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- 9 **VENDOR** Adams et al. (Meta), "Automating Low-Risk Code Review at Meta: RADAR, Risk Calibration, and Review Efficiency," arXiv:2605.30208 — Meta-reported production metrics. v2 2026-06-12.
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- 10 **VENDOR** Spotify Engineering, on Fleetshift's deterministic auto-merged transforms (no model involved) and human review of every LLM PR. Jun 2026 · accessed 2026-08-27.
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- 11 **ANECDOTE** Thomas Ptacek, "My AI Skeptic Friends Are All Nuts," fly.io blog. Stake: employer sells production infrastructure, not agents. 2 Jun 2025.
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- 12 **VENDOR** Cursor Terms of Service §1.7, "Auto-Code Execution" (contract text). Accessed 2026-08-30.
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- 13 **STUDY** This project, `research/verification-infrastructure.md` — the spine tested; eval decay (Cline workflow deletion, primary git history); the one verification-floor artifact. Closed 2026-08-28.
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- 14 **STUDY** This project, `research/verification-observability.md` — the central negative: no authorship-keyed gating, canarying or auto-revert published anywhere. Closed 2026-08-28.
- 
- 15 **VENDOR** Anthropic-published telemetry: ~93%→97% permission-prompt approval; human review catching 13.6% of dangerous commands vs 89% automated (n=1,053). 2026.
- 
- 16 **STUDY** secunet, on agents defeating proof obligations ( `pragma Assume` , then `SPARK_Mode => Off` ), arXiv:2607.14340. 2026.
- 
- 17 **STUDY** NIST SP 800-218A §2 Scope — sentence verified verbatim from the nvlpubs PDF; scope caveat: the profile governs secure development of AI models. Jul 2024 · re-verified 2026-08-31.
- 
- 18 **STUDY** CISA/OMB Secure Software Development Attestation Common Form (OMB #1670-0052) — wording verified verbatim from the form PDF. 2024 · re-verified 2026-08-31.
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- 19 **VENDOR** GitHub Terms of Service §J.4 (contract text). Effective 2026-04-27.
- 
- 20 **STUDY** US Copyright Office, "Copyright and Artificial Intelligence, Part 2: Copyrightability" (official guidance). 2025.
-

- 21 **STUDY** CMORG, "Firm Guidance For Frontier AI," Final v1.0, TLP:CLEAR — read in full from the public PDF. 4 Jun 2026 · read 2026-08-31.
- 
- 22 **VENDOR** This project, [research/ai-native-sdlc-playbook.md](#) — digest of Anthropic Academy's *The AI-Native SDLC Playbook* (14 lessons; © Anthropic, reference copy in [reference/](#) ). The vendor's recommended practice for its own product: indicators defined for every play, values reported for none — nothing here is an outcome. Cited on this page: L8 (the anti-suppression hook), L11 (managed settings; hooks as approval gates), L12 (the withheld production gate; rehearsed rollback), L13 (response bands; the pre-approved runbook). Retrieved and digested 2026-09-02.
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# Glossary

The vocabulary these documents are written in. Definitions are the project's ruling usage, established against primary sources; where a common term is avoided, the reason is given.

Vocabulary as of 2026-08-31    Taxonomy: ADR-0004    Evidence in `research/vocabulary-and-disciplines.md`

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## 01 The five archetypes

An **archetype** is a grouping of teams that work with AI in the same way — named by the team's configured *ceiling* for model-generated changes, and ordered by what the agent's output can reach with no human action in between.

### Autocomplete

The agent's output reaches the open file; a human accepts every edit. The spectrum's fixed low extreme: inline AI edits with nothing else about the process changing.

### Workspace

The agent holds file-write and shell tool surfaces and acts across the working tree, locally. No push credential — nothing leaves the machine without a human.

### Contributor

The agent holds a push credential; its output enters the shared repository, and a human merges every change. The large middle of observed practice.

### Committer

A defined class of model-generated changes holds merge rights and reaches trunk unread; a human still gates release.

### Operator

Nothing human stands between the agent's change and running production — by continuous deployment from trunk or agent-held deploy credentials. Reachable per change-class; observed as no team's ceiling.

## 02 The spectrum's own terms

### Archetype

A grouping of teams that work with AI in the same way, defined by what the human stops doing and what engineered verification replaces them with. An archetype is a team's configured *ceiling* — not what any individual change receives. Archetypes are kinds, not ranks: none is the goal state.

**AVOID** tier and level (imply ranking) · framework (means React or Django to this audience) · model (reads as *language model*) · stage, maturity level (imply progression)

### Ceiling

The maximum delegation a team's settings and policies permit **for model-generated changes** — what an administrator flips, and what vendor terms key liability to. A property of the configured scope where the settings live (an organisation, a repository, a path); "team" is shorthand for a team at a scope. Deterministic automation with no model involved sits outside the spectrum entirely.

**AVOID** describing a team by its typical or modal change — that is unmeasurable from outside and drifts as volume shifts between change types

### Spectrum

The set of archetypes ordered by containment ceiling — what the agent is permitted to reach. A boundary is crossed when the agent's output can reach the next surface (open file, working tree, shared repository, trunk, production) with no human action in between; the usual mechanism is a credential or tool surface, held by the agent or pre-authorised on its behalf.

**AVOID** maturity model, adoption curve, ladder (all imply higher is better) · write "further along the spectrum", never "higher up"

### The independence rule

Containment and human oversight move independently, and the spectrum orders only containment. A team may grant an agent very little reach and supervise none of it, or grant it production reach and gate every action. Every archetype document therefore states three things separately: what the agent can reach, what a human still gates, and what verification must replace them.

**AVOID** writing or implying that further along the spectrum means less supervised — that conflation is the specific error these documents exist to break

### Posture

The set of beliefs and constraints — about trust, risk and what AI should touch — that lands a team in a particular archetype. Each archetype page closes with the posture that fits it.

**AVOID** maturity, sophistication, readiness

## 03 **Autonomy, oversight, containment**

### **Autonomy**

A per-action property — what this agent may do without approval, in this context. Described by naming what the human approves and when.

**AVOID** "L3 autonomy" or any bare level reference — no standards body has published a scale, and the two leading proposals contradict each other

### **Human oversight**

What a human actually gates, named specifically. The register regulators use.

**AVOID** "human-in-the-loop" as a reassurance claim: the vendor holding the telemetry reports ~93% rising to 97% of permission prompts approved, and human review catching 13.6% of dangerous commands against 89% for automated checking — the mechanism does not imply the outcome

### **Containment**

Restricting what an agent *can* do — sandboxes, egress controls, least-privilege tool scopes — rather than approving what it does. The field's structural answer to the limits of supervision, and the property the spectrum orders.

**AVOID** "guardrails" as a load-bearing noun; name the mechanism (validation, allowlist, sandbox, classifier, tripwire, permission rule, hook)

### **Verification**

The engineered machinery that establishes a change is acceptable when a human has not read it: tests, gates, static analysis, canaries, rollback, evals.

**AVOID** using "review" for both human reading and automated checking — they are the two things these documents most need to keep apart

### **Lethal trifecta**

The combination of private data access, exposure to untrusted content, and the ability to communicate externally — the configuration under which prompt injection becomes exfiltration.

### **Perception gap**

The measured difference between how much faster developers believe AI makes them and how much faster they measurably are — consistently large, and consistently in the direction of belief exceeding measurement.

**AVOID** presenting it as proof that AI makes developers slower; the gap is well-evidenced, the slowdown is not

## 04 The disciplines

### Context engineering

The discipline of assembling what an agent sees: instructions, retrieved code, tool results, history, and what is deliberately excluded. Coined by Tobi Lütke (19 Jun 2025), amplified by Andrej Karpathy (25 Jun 2025).

**AVOID** "prompt engineering" as the name of the discipline — it is now a subset, meaning the wording of instructions specifically

### Agent harness

Everything around the model that makes it an agent: the agent loop, tool surfaces, permission modes, hooks, sandboxes, and feedback loops.

**AVOID** bare "harness" where eval harnesses are also discussed

### Harness engineering

The discipline of designing an agent harness. Named by Mitchell Hashimoto (5 Feb 2026) and industry vocabulary within months; measurably load-bearing — harness choice moves benchmark results about as much as model choice.

**AVOID** "scaffolding" as a distinction to build an argument on — it is the 2024 name for the same thing

### Eval harness

The evaluation infrastructure that runs a benchmark and scores results. A different thing from an agent harness; conflating them corrupts benchmark numbers.

### Agent orchestration

The discipline of structuring work across multiple agent steps or agents: subagent fan-out, pipelines, gates, checkpoints, handoffs.

**AVOID** "graph engineering" — not a term of art anywhere in this field, and it collides with the unrelated "knowledge graph engineering"

### Flow engineering

The narrower practice of encoding control flow explicitly rather than leaving the sequence to the agent's judgement. A technique within agent orchestration, not a synonym for it.

### Evals

Task-specific, purpose-built measurements of whether a system does acceptable work. Distinct in practice from public benchmarks.

**AVOID** treating evals and benchmarks as interchangeable

## Agentic engineering

Professional software development where agents do substantial work under engineered constraint — the current name for the practice the later archetypes describe.

**AVOID** "vibe coding" for disciplined work (it means prototypes where the diffs go unread) · "vibe engineering" (retired by its own author)

## 05 How these documents cite

### Evidence tier

The classification every cited figure carries, stated with the claim itself — never as a blanket disclaimer: survey hard survey data · study controlled study or verified primary artifact · vendor vendor-reported metric · anecdote practitioner account.

### Joint measurement

An agent benchmark score, which reflects model and harness together. Always reported as "model X + harness Y resolves N%", never as a property of the model alone.

**AVOID** any bare agent benchmark score; SWE-bench Verified scores entirely — deprecated by their own publisher in February 2026 for training contamination

## 06 A note on external scales

No standards body has published an autonomy-levels scale — verified against NIST AI RMF, ISO/IEC 22989, IEEE P3394, SAE and the EU AI Act — which is why this project defines its own archetypes and says so.

### THE MILLS REVIEW IS A DIFFERENT AXIS

The FCA-commissioned **Mills Review** (Sheldon Mills, published 6 July 2026) contains an L1–L5 autonomy spectrum whose worked examples are about writing code. It orders *autonomy*; this spectrum orders *containment* — independent axes. The one rough correspondence: its L4 ("AI writes, tests and stages code; engineer approves each release") sits near the Committer archetype. Do not map the rest, and never cite its levels bare — a commissioned review is not a standard.